WARD: A Vision-Only Framework for Water-Surface Awareness and Risk Detection

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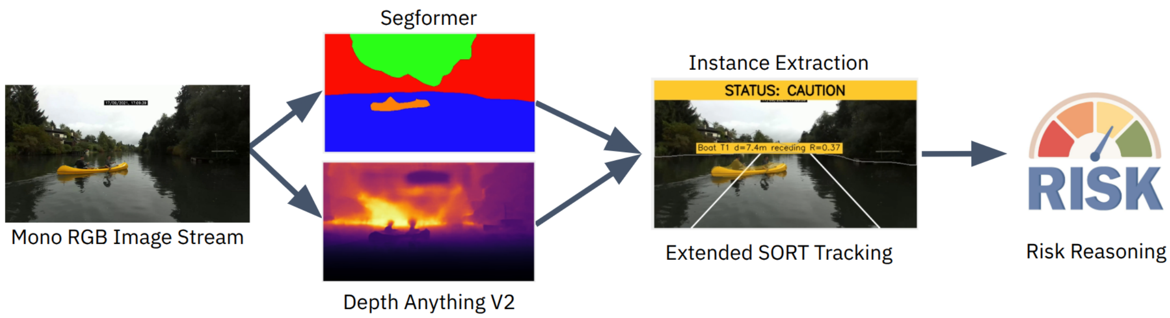

Fig. 1: Overview of the proposed framework. Monocular RGB frames are processed by SegFormer and Depth Anything V2 to extract semantic and depth cues, which are then fused to generate actionable safety warnings.

***Abstract*—Safe navigation for Autonomous Surface Vehicles (ASVs) requires both accurate obstacle detection and actionable risk assessment. We propose WARD (Water-surface Awareness and Risk Detection), a vision-only, risk-aware framework that can output safety assessments for low-cost ASVs. WARD fuses SegFormer semantic segmentation with radar-calibrated Depth Anything V2 monocular depth using a novel continuous collision-corridor risk model. The risk model reasons jointly about proximity, lateral offset, and semantic hazards to produce human-interpretable safety alerts. An extended SORT tracker ensures stable Time-to-Collision predictions by filtering monocular depth noise. Operating at 13.7 FPS on a consumer-grade GPU, WARD provides reliable, discrete safety warnings (SAFE, CAUTION, IMMEDIATE) for resource-constrained ASVs.**

I. Introduction

Autonomous Surface Vehicles (ASVs) are increasingly deployed in unstructured maritime environments such as busy ports and marinas. These domains present unique perception challenges, including dynamic water reflections, waves, and low-profile hazards. While high-end platforms rely on active sensors like marine radar or LiDAR, small-scale ASVs demand low-cost, vision-only solutions that operate reliably under strict power and hardware constraints.

In safety-critical navigation, a gap exists between perception accuracy and operational safety: high accuracy in perception modules, such as object detection and segmentation, does not inherently translate into safe navigation. The fundamental question for an ASV is not whether an obstacle is perfectly segmented, but a more urgent one: *"Does this obstacle pose an immediate collision risk?"* A vision system that returns raw detections without this risk-aware context can overwhelm the controller with data while failing to highlight the most pressing threats, such as treating a distant, harmless buoy and a swimmer in the immediate path as equally salient.

To bridge this gap, we propose WARD (Water-surface Awareness and Risk Detection), a vision-only (RGB), risk-aware framework that fuses SegFormer [1] semantic segmentation with radar-calibrated Depth Anything V2 [2] monocular depth, as illustrated in Fig. 1. WARD introduces a continuous collision-corridor risk model to reason jointly about proximity, lateral offset, closing velocity, and semantic hazard. By delivering actionable, real-time safety assessments (SAFE, CAUTION, and IMMEDIATE) from a single RGB stream, WARD offers a practical baseline for resource-constrained ASVs operating in dynamic environments.

II. Technical Approach

WARD's modular pipeline processes RGB frames through parallel SegFormer and Depth Anything V2 modules. These outputs are fused via our continuous collision-corridor risk model to generate per-obstacle risk scores. To ensure stability, an extended SORT [3] tracker smooths these scores before mapping to discrete warning levels.

A. Segmentation and Calibrated Depth

We adopt SegFormer (MiT-B0) trained on the LaRS [4] dataset as our segmentation backbone. To enable instance separation from a semantic-only backbone, we extend it with a dual-head design: a 9-class semantic head and an auxiliary obstacle-boundary head, the latter used to split adjacent same-class instances during connected-component extraction. Depth is estimated using the Metric-Outdoor variant of Depth Anything V2, fine-tuned from the foundation model on synthetic Virtual KITTI data [5]. To correct scale bias from road-centric pre-training without retraining on scarce maritime data, we apply a single global scaling factor $s_{cal} = 0.4$ calibrated once against radar ground truth from the WaterScenes dataset [6] according to $d_{\text{cal}} = s_{cal} \cdot d_{\text{raw}}$, where d_{raw} is the raw output from the Depth Anything V2 model and d_{cal} represents the calibrated metric depth in meters. This adjustment restores near-metric accuracy in the critical 0–20 m close-up range, which is the primary zone for maritime collision warning.

B. Risk Model and Extended Tracking

Based on the Responsibility-Sensitive Safety (RSS) framework [7], we define a virtual forward collision corridor of width $W_{\text{boat}} + 2M_{\text{lat}}$ centered on the ASV heading, where W_{boat} is the ASV beam width and M_{lat} is the lateral safety margin. For an obstacle with forward distance d_{fwd} and lateral excess e_{lat} outside this corridor, we first compute a static risk baseline:

$$R_{\text{base}} = w_{\text{class}} \cdot \exp(-d_{\text{fwd}}/D_{\text{safe}}) \cdot \exp(-e_{\text{lat}}/L_{\text{safe}}), \quad (1)$$

where w_{class} is the semantic hazard weight, D_{safe} is the forward decay constant, and L_{safe} is the lateral decay constant. This baseline is amplified by a closing-velocity factor:

$$R = R_{\text{base}} \cdot \big(1 + \alpha_v \cdot \min(v_{\text{closing}}/V_{\text{ref}},\ 2)\big), \quad (2)$$

where v_{closing} is the Kalman-filtered closing velocity, V_{ref} is the reference speed, and α_v is the amplification factor. The continuous score R is mapped to three discrete warning levels, SAFE ($R < 0.15$), CAUTION ($0.15 \leq R < 0.50$), and IMMEDIATE($R \geq 0.50$), via threshold hysteresis with separate enter/exit values to suppress noise-induced flicker.

To mitigate monocular depth noise, we extend the SORT tracker to a 9-D Kalman state $\mathbf{x} = [c_x, c_y, s, r, d, \dot{c}_x, \dot{c}_y, \dot{s}, \dot{d}]^\top$, where (c_x, c_y), s, and r denote the bounding-box center, scale (area), and aspect ratio; d is the calibrated metric depth; and dotted variables represent their respective rates of change. The filter optimally balances depth observation noise against a constant-velocity process model. This yields smoothed estimates of the closing velocity, $v_{\text{closing}} = -\dot{d} \cdot f_{\text{fps}}$, and the Time-to-Collision, $TTC = d/v_{\text{closing}}$, where f_{fps} is the system frame rate. These parameters feed directly into the risk model in Eq. (2) without requiring post-hoc exponential moving averages or windowing.

III. Results

We evaluated the segmentation backbone on the LaRS validation split. SegFormer achieves an obstacle recall of 96.51%, substantially outperforming a DeepLab baseline (79.83% recall) at comparable cost. This high recall is critical for safety-oriented ASV operation, where missed low-profile hazards such as buoys, swimmers, or floating debris can be catastrophic. For monocular depth, the evaluation was conducted using the radar data from the WaterScenes dataset, where the Base model provides a mean absolute error (MAE) of 4.04 m within the critical 0–20 m range. While the Large variant offers lower error (3.22 m), its high latency is prohibitive for real-time edge navigation.

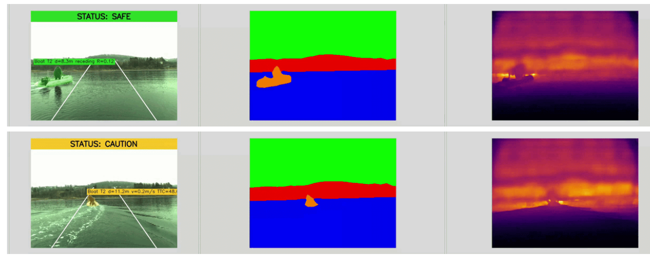

Fig. 2: Qualitative evaluation on a MODD sequence. As the tracked boat enters the virtual collision corridor, the system transitions from SAFE (top) to CAUTION (bottom). Each row shows the annotated RGB frame (left), semantic segmentation (center), and monocular depth map (right).

The integrated pipeline was evaluated on real-world maritime sequences from the MODD [8] dataset. As demonstrated by the qualitative results in Fig. 2, the system reliably distinguishes between stable and escalating risk states. The system maintains these transitions with high temporal stability; the warning distribution (62% SAFE, 33% CAUTION, 5% IMMEDIATE) remained consistent with the obstacle dynamics, with no observed warning-level flicker.

In terms of computational efficiency, the total system latency is 73.0 ms, sustaining approximately 13.7 FPS on a NVIDIA RTX 4060 GPU. Depth estimation is the primary bottleneck (38.7 ms), followed by segmentation (26.9 ms) and risk reasoning (7.4 ms). This low overhead for risk reasoning leaves significant headroom for future integration of higher-rate sensors or more complex fusion logic.

IV. Conclusion

We presented WARD, a lightweight, vision-only framework that bridges raw perception and actionable risk assessment. By delivering stable, human-interpretable warnings without active ranging hardware, WARD enables safe navigation for low-cost ASVs. Future work targets field deployment, 360° multi-camera coverage, and closed-loop risk-aware control.

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